1 FIG-212 — From-scratch reproduction threshold onset

Group: MD-FROM-SCRATCH-FAILURE-ATTRIBUTION **Evidence chain:** manifested **Status:** operational instability threshold across three seeds; no Dxy for 8 of 9 branches

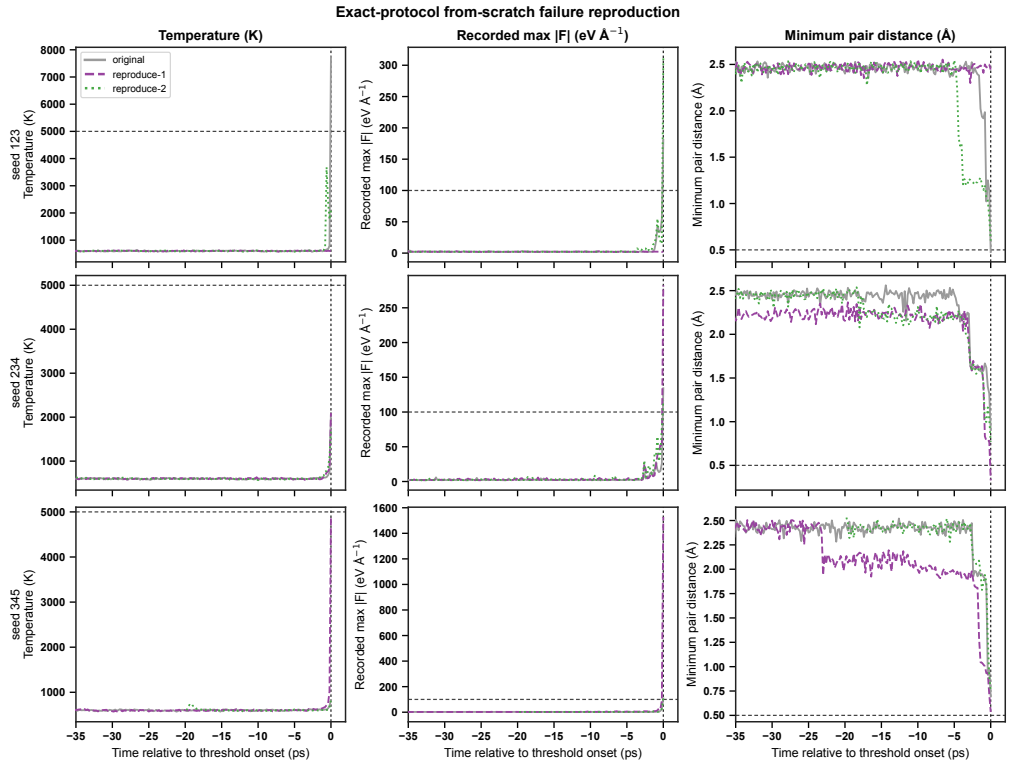


Figure 1: A–I, three-seed time courses for the original and two reproduction branches. The horizontal line marks the declared operational instability threshold; the vertical marker indicates the first saved crossing when present. One branch completed the 1000 ps target without crossing the threshold, while the remaining branches crossed it at different times. This is a diagnostic of the declared gate, not a claim of a thermodynamic phase transition.

Results text draft. Eight of nine audited branches crossed the operational instability threshold before the 1000 ps target; one reproduce-1 seed-123 branch reached 1000 ps without a threshold crossing. First crossings ranged from 19.8 ps to 795.6 ps among the branches with an onset.

Discussion and boundary. The panel establishes a reproducible dynamic-stability failure boundary for the tested from-scratch campaign. It does not establish a microscopic cause, calibrated uncertainty, a converged transport coefficient, or a physical transition. Unstable branches remain labelled "unstable – no Dxy".

Article placement. Proposed article location: Supplementary Information / from-scratch instability and failure-attribution audit; not yet a main-text transport claim.

Reviewer linkage: overall status PENDING.

- **REV-C9 (PENDING)**: Resolve the from-scratch dynamic-stability boundary before reporting transport quantities. *Completed:* Nine three-seed branches were audited with an explicit operational instability threshold. *Still required:* Independently close the scientific stability and transport-acceptance gates before assigning Dxy or using this diagnostic as a transport result.

Evidence pointers.

- Data/processed: `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/metrics_all.csv`
- Data/processed: `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/replica_ledger.csv`

- Plot/processing script: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/plot_failure_attribution_all.py
- Plot Atlas and figure manifest: /Users/alina/Documents/Codex/plot_library; /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/figures/figure_manifest.json
- Asset SHA256: 1e01a2bcd7d70e62e02d000ecccb29d236a197e86fbf44453bb4b756702737cc

Machine-collected evidence candidates (generated from the analysis manifest; do not edit this file).

- **Diagnostic report:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/RESULTS_DISCUSSION.md
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/run_summary.csv
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/plot_failure_attribution_all.py
- **Plot Atlas source:** /Users/alina/Documents/Codex/plot_library

Audit gap / next action: The threshold is an operational analysis gate. Complete the independent stability and transport acceptance checks before promoting any branch to a Dxy result or causal failure statement.